GRADUATION PROJECT FINAL COMPLETION AUDIT & SQUAD CONTRIBUTION REPORT (100 / 100 PTS)

SOC Analyst & Incident Responder Bootcamp Capstone

Squad Members: Mohammad Bustami • Saif Asender • Mohammad Alodeh | GitHub: `https://github.com/MB-Cyber1/Wazuh-Capstone`

OVERALL STATUS	3-MEMBER SQUAD	SIEM & LAB STACK	GITHUB REPOSITORY
100% Completed 14 / 14 Sub-tasks Verified (100 Pts)	**M. Bustami, S. Asender, M. Alodeh** Full Separation of Duties & Peer Review	**Wazuh v4.14.9 (Docker)** 3 VMs Live on VMnet1 (192.168.99.0/24)	**MB-Cyber1/Wazuh-Capstone** 56 Deliverables Pushed to main

EXECUTIVE COMPLETION SUMMARY — 100% OF PART 1 (60 PTS) & PART 2 (40 PTS) DELIVERED

All mandatory requirements from the Graduation Project brief are **100% completed, verified on the live 3-VM Wazuh SIEM lab, signed off by all 3 squad members (Mohammad Bustami, Saif Asender, and Mohammad Alodeh), and pushed to GitHub (`https://github.com/MB-Cyber1/Wazuh-Capstone`)**. Every team member authored ≥ 3 detection rules, peer-reviewed teammates' Detection Cards, triaged dedicated incident tickets under strict red/blue separation of duties, and co-authored the Part 2 Incident Readiness Pack.

1. MASTER RUBRIC COMPLETION & SQUAD OWNERSHIP MATRIX (100 POINTS TOTAL)

TASK	REQUIREMENT & WEIGHT	STATUS	COMPLETED DELIVERABLES & VERIFICATION PROOF (IN GITHUB REPO)	SQUAD LEAD & CONTRIBUTORS
1.1	**Build the Lab & Log Pipeline** (6 pts)	100% DONE	• `SOC-Wazuh` (`192.168.99.10`): Docker single-node stack (`manager` , `indexer` , `dashboard` v4.14.9). • **Source 1**: `Win10` (`192.168.99.30` , Agent `002`) Sysmon v15.15 & EventChannel (`0.57s` delay). • **Source 2 & 3**: `ubuntu-01` (`192.168.99.40` , Agent `001`) Linux `auditd` (`0.83s`) & `nginx` logs (`0.45s`). • Proof in `evidence/PIPELINE_PROOF.md` .	**Leads:** Mohammad Bustami & Saif Asender **Support:** Mohammad Alodeh
1.2	**10 Rules Fire with Evidence** (10 pts)	100% DONE	• All 10 detection rules + correlation rule `100200` (*Brute force followed by logon success*) deployed in `detections/wazuh/local_rules.xml` . • All 10 attack triggers executed live on `Win10` & `ubuntu-01` ; raw alert JSONs extracted into `evidence/alerts_evidence.json` and `evidence/RULE_EVIDENCE_SUMMARY.md` .	**Mohammad Bustami:** 4 Rules **Saif Asender:** 3 Rules **Mohammad Alodeh:** 3 Rules
1.3	**Detection Logic, Sigma & ATT&CK v19** (12 pts)	100% DONE	• Behavior-based PCRE2 & syscall logic across **6 ATT&CK v19 tactics** (including v19 `Stealth TA0005` & `Defense Impairment TA0112`) and **2 Linux rules** (`100202` *SSH authorized_keys modified*, `100203` *Linux cron job change*) + correlation rule `100200` . • 10 Sigma v2 files pass `sigma check` (0 errors); translation notes in `detections/CONVERSION_LOG.md` . • Navigator layer & gap note in `detections/navigator/` .	**All 3 Members** (Each authored Sigma + XML for their assigned rules; Navigator by M. Bustami)
1.4	**Cards, Benign Testing, Tuning & Dashboard** (9 pts)	100% DONE	• 10 one-page Detection Cards in `detections/cards/card_01..10.md` peer-reviewed across all 3 members. • 10 benign tests executed post-tuning with **0 false positives**; tuning counts in `detections/tuning/TUNING_REPORT.md` . • 6-panel custom Tier-1 Dashboard in `detections/dashboard/` .	**Tuning Lead:** Mohammad Alodeh **Dashboard Lead:** Mohammad Bustami **Cards & Peer Review:** All 3
1.5	**Alert Triage, Escalation & Containment** (14 pts)	100% DONE	• 4 complete 10-item tickets in `triage/ticket_01..04.md` (Incident A TP chain & Incident B LSASS FP). • Six-field escalation call (`triage/escalation_call_incident_a.md`), firewall isolation, VMware forensic snapshot (`SOC-Win10-Snapshot1.vmsn` + 4 GB `.vmem`) & SHA-256 hashes in `evidence/SHA256SUMS.txt` .	**Red-Team Attacker:** Saif Asender **Incident A Analyst:** Mohammad Bustami **Incident B FP Analyst:** Mohammad Alodeh
1.6	**Incident A Report** (9 pts)	100% DONE	• Full Incident A investigation report in `triage/INCIDENT_A_REPORT.md` with UTC chronological timeline, Mermaid kill-chain diagram, root cause, containment/eradication/recovery, ATT&CK v19 mapping, and lessons learned.	**Lead Author:** Mohammad Bustami **Co-Authors:** Saif Asender & Mohammad Alodeh
2.1	**Org Profile & Risk Register** (6 pts)	100% DONE	• `readiness/01_org_profile_and_risk_register.md` : Fictional healthcare provider (**Medira Health Systems**), MTD/RTO/RPO limits, lab-to-org mapping, **12-row Risk Register** (with explicit `No Coverage` rows), and Top 3 risk deep dive.	**Lead Author:** Mohammad Bustami **Reviewer:** Saif Asender
2.2	**Severity Matrix & Card Alignment** (4 pts)	100% DONE	• `readiness/02_severity_classification_matrix.md` : P1–P4 matrix with **12 concrete org-specific examples** (3 per level), SLAs, update intervals, declare/downgrade authority, and 5 updated Part 1 Detection Cards.	**Lead Author:** Mohammad Alodeh **Reviewer:** Mohammad Bustami
2.3	**Escalation Tree & National Reporting** (4 pts)	100% DONE	• `readiness/03_escalation_tree_and_contacts.md` : Business-hours vs. after-hours call order, **10-minute no-answer fallback** policy, six-field call structure, 24/7 contact roster with OOB Signal channel, and SDAIA/NCA reporting.	**Lead Author:** Saif Asender **Reviewer:** Mohammad Alodeh
2.4	**Three IR Playbooks & Chain of Custody** (5 pts)	100% DONE	• `readiness/04_ir_playbooks.md` : 3 concise playbooks (`PB-01` LSASS/PowerShell/Log Wipe, `PB-02` Brute Force/RDP, `PB-03` Linux SSH Key/Cron) with Mermaid decision flowcharts and completed Chain-of-Custody form.	**PB-01:** Mohammad Bustami **PB-02:** Mohammad Alodeh **PB-03:** Saif Asender

TASK	REQUIREMENT & WEIGHT	STATUS	COMPLETED DELIVERABLES & VERIFICATION PROOF (IN GITHUB REPO)	SQUAD LEAD & CONTRIBUTORS
2.5	**Playbook Testing & Cold Test** (6 pts)	**100% DONE**	• `readiness/05_playbook_testing_sheet.md` : Internal timed runs (`2m 10s` , `1m 45s` , `2m 05s`), 5 friction points fixed, plus completed **External Student Cold-Test Log Sheet** (`PB-02` completed in `2m 38s ≤ 3m 30s` max, 1 question, PASS).	**Internal Run:** All 3 Members **Cold-Test Observers:** Mohammad Alodeh & Saif Asender
2.6	**Communication Plan** (3 pts)	**100% DONE**	• `readiness/06_communication_plan.md` : Spokesperson & approval matrix plus ready-to-send P1 templates for all 5 audiences: Executive/Board, Internal Staff, Regulator (72h PDPL/NCA), Customers/Patients, and Media Holding Statement.	**Lead Author:** Mohammad Alodeh **Reviewer:** Saif Asender
2.7	**Tabletop Exercise (TTX) Package** (7 pts)	**100% DONE**	• `readiness/07_tabletop_exercise_package.md` : 45-min scenario ("*Operation Silent Ward*") with 6 injects (including FP curveball, log-wipe curveball, and media/clinical director pressure inject), AAR action items, and completed Observer Log timings.	**Scenario & Run:** Mohammad Bustami, Saif Asender & Mohammad Alodeh
2.8	**Executive Briefing Document** (5 pts)	**100% DONE**	• `readiness/08_executive_briefing.md` : One-page executive summary, readiness metrics, top 3 residual risks, 30/60/90-day roadmap, cited cost-of-inaction figures (IBM 2024 Healthcare $9.77M & Verizon DBIR 2024), and 10-slide outline.	**Leads:** Mohammad Bustami & Mohammad Alodeh

2. WHAT EACH TEAM MEMBER DID — DETAILED INDIVIDUAL CONTRIBUTION BREAKDOWN

1. Mohammad Bustami

SIEM ARCHITECT, DETECTION LEAD & INCIDENT A BLUE-TEAM LEAD

- **Lab & SIEM Stack (1.1):** Deployed `SOC-Wazuh` (`192.168.99.10`) Docker v4.14.9 stack and integrated Windows EventChannel/Sysmon on `Win10` (`0.57s` latency).
- **4 Detection Rules + Correlation (1.2 & 1.3):** Authored Sigma v2 + Wazuh XML & Detection Cards for:
  - `100005` : **Suspicious PowerShell (encoded command or download cradle)** (`T1059.001`)
  - `100007 + 100200` : **5 failed logons in 2 min → Brute force followed by logon success** (`T1110 / T1078`)
  - `100201` : **LSASS memory access (credential dumping)** (`T1003.001`)
  - `100207` : **Windows Security/System event log cleared** (`T1685.005` , ATT&CK v19 `TA0112`)
- **Peer Review (1.4):** Reviewed 4 cards authored by teammates: `100013` (*LOLBin download*), `100202` (*Linux SSH authorized_keys modified*), `100204` (*External RDP login*), and `100206` (*Registry Run/RunOnce persistence*).
- **Dashboard & Navigator (1.3 & 1.4):** Built the 6-panel custom Tier-1 Dashboard (`soc_capstone_tier1_dashboard.ndjson`) and ATT&CK v19 Navigator layer + Gap Note.
- **Incident A Blue-Team Lead (1.5 & 1.6):** Triaged **Ticket #1 (`SOC-2026-INC-A01`)**, delivered the 6-Field Escalation Call, executed host firewall isolation & VMware snapshot hashing (`SHA256SUMS.txt`), and authored `INCIDENT_A_REPORT.md` .
- **Part 2 Readiness Pack (2.1, 2.4, 2.7, 2.8):** Authored Org Profile & 12-row Risk Register (`2.1`), Playbook `PB-01` (`2.4`), TTX Injects 3–5 response (`2.7`), and co-authored Executive Briefing (`2.8`).

2. Saif Asender

LINUX/WINDOWS TELEMETRY ENGINEER & RED-TEAM OPERATOR

- **Linux & Web Pipeline (1.1):** Configured `ubuntu-01` (`192.168.99.40` , Agent `001`) with `auditd` (`Neo23x0` + `10-soc-capstone.rules` , `0.83s` delay) and `nginx` web logs (`0.45s` delay), resolving the `audit.log` read feedback loop.
- **3 Detection Rules (1.2 & 1.3):** Authored Sigma v2 + Wazuh XML & Detection Cards for:
  - `100013` : **LOLBin file download (certutil, bitsadmin, mshta, regsvr32)** (`T1105`)
  - `100203` : **Linux cron job change (/usr/bin/crontab)** (`T1053.003`)
  - `100202` : **Linux SSH authorized_keys modified** (`T1098.004`)
- **Peer Review (1.4):** Reviewed 3 cards authored by teammates: `100005` (*Suspicious PowerShell*), `100201` (*LSASS memory access*), and `100205` (*Suspicious rundll32 proxy execution*).
- **Incident A Red-Team Operator & Ticket #4 Lead (1.5):** Executed the multi-stage attack chain on `Win10` & `ubuntu-01` for blind Blue-Team triage, and served as Lead Blue-Team Analyst for **Ticket #4 (`SOC-2026-INC-D04` — Windows Security Event Log Cleared `1102`)**.
- **Part 2 Readiness Pack (2.3, 2.4, 2.5, 2.7):** Authored Escalation Tree, 10-min fallback & SDAIA/NCA reporting (`2.3`), Linux Playbook `PB-03` (`2.4`), co-observed the External Cold Playbook Test (`2.5`), and led TTX Injects 1 & 5 (`2.7`).

3. Mohammad Alodeh

TUNING & FALSE-POSITIVE LEAD (INCIDENT B) & READINESS LEAD

- **3 Detection Rules (1.2 & 1.3):** Authored Sigma v2 + Wazuh XML & Detection Cards for:
  - `100204` : **Remote Desktop (RDP) login from external IP (Logon Type 10)** (`T1021.001`)
  - `100205` : **Suspicious rundll32.exe proxy execution** (`T1218.011` , ATT&CK v19 `Stealth TA0005`)
  - `100206` : **Registry Run/RunOnce key modified via SetValue** (`T1547.001`)
- **Peer Review (1.4):** Reviewed 3 cards authored by teammates: `100007 + 100200` (*Brute force → logon success correlation*), `100203` (*Linux cron job change*), and `100207` (*Windows Security/System log cleared*).
- **Benign Testing & Rule Tuning Lead (1.3 & 1.4):** Led root-cause analysis and PCRE2 tuning in `TUNING_REPORT.md` , eliminating all 26 false positives on Rule `100201` (*LSASS memory access* from `svchost.exe` / `MsMpEng.exe` `0x1000` queries) and verifying 0 false positives across all 10 benign tests.
- **Incident B FP Lead & Ticket #3 Lead (1.5):** Triaged **Ticket #2 (`SOC-2026-INC-B02` — Confirmed False Positive on LSASS Rule 100201)** and **Ticket #3 (`SOC-2026-INC-C03` — Linux SSH & Cron Persistence Rules 100202 & 100203)**.
- **Part 2 Readiness Pack (2.2, 2.4, 2.5, 2.6, 2.7, 2.8):** Authored Severity Matrix & 5-Card SLA alignment (`2.2`), Playbook `PB-02` (`2.4`), Cold Test timing log (`2.5`), 5-Audience Crisis Communication Plan (`2.6`), TTX Injects 2 & 6 response (`2.7`), and co-authored Executive Briefing (`2.8`).

2.1 Rule-by-Rule Plain-Text Detection Summary, Ownership, Peer Review & Live Firing Matrix (Sub-tasks 1.2, 1.3, 1.4)

ID	DETECTS (WHAT IT CATCHES IN PLAIN TEXT)	ATT&CK V19	LEVEL	OWNER & REVIEWER	LIVE FIRED TIMESTAMP (UTC) & VERIFIED TRIGGER / LOGIC
`100005`	**Suspicious PowerShell (encoded, download)**	`T1059.001` Execution (`TA0002`)	Lvl 10 **P2**	**M. Bustami** *Rev: S. Asender*	`2026-10-19T18:43:30.754Z` — Fired on `powershell.exe -NoProfile -enc dwBoAG8AYQBtAGkA` . Matches `-enc` , `FromBase64String` , `DownloadString` , `iwr` , `iex` .
`100007`	**5 failed logons in 2 min (Windows brute force)**	`T1110` Cred Access (`TA0006`)	Lvl 10 **P2**	**M. Bustami** *Rev: M. Alodeh*	`2026-10-19T18:43:36.474Z` — Stateful threshold rule (`if_matched_sid: 60122` , `frequency="5"` , `timeframe="120"` , `same_field: targetUserName`).
`100200` *(Corr.)*	**Brute force followed by logon success (Correlation)**	`T1110 + T1078` Cred Access (`TA0006`)	Lvl 13 **P1**	**M. Bustami** *Rev: M. Alodeh*	`2026-10-19T18:43:40.196Z` — **Correlation Rule**: Matches Event `4624` when preceded by Rule `100007` for the same `targetUserName` within 300s.

ID	DETECTS (WHAT IT CATCHES IN PLAIN TEXT)	ATT&CK V19	LEVEL	OWNER & REVIEWER	LIVE FIRED TIMESTAMP (UTC) & VERIFIED TRIGGER / LOGIC
`100201`	**LSASS memory access (credential dumping)**	`T1003.001` Cred Access (`TA0006`)	Lvl 14 **P1**	**M. Bustami** *Rev: S. Asender*	`2026-10-19T18:52:15.234Z` — Detects non-system processes accessing `lsass.exe` (Sysmon EID 10). Tuned PCRE2 lookahead (0 false positives).
`100013`	**LOLBin download (certutil, bitsadmin, mshta)**	`T1105` Command & Control (`TA0011`)	Lvl 10 **P2**	**S. Asender** *Rev: M. Bustami*	`2026-10-19T18:49:42.921Z` — Detects file downloads via `certutil` , `bitsadmin` , `mshta` , or `regsvr32` . Fired on `mshta http://192.168.99.40/index.html` .
`100203`	**Linux cron job change (/usr/bin/crontab)**	`T1053.003` Persistence (`TA0003`)	Lvl 8 **P3/P2**	**S. Asender** *Rev: M. Alodeh*	`2026-10-19T17:59:14.292Z` — Linux rule #1: Matches `audit.exe=/usr/bin/crontab` with `audit.syscall=59` (`execve`) on `ubuntu-01` .
`100202`	**Linux SSH authorized_keys modified**	`T1098.004` Persistence (`TA0003`)	Lvl 10 **P2**	**S. Asender** *Rev: M. Bustami*	`2026-10-19T17:57:36.152Z` — Linux rule #2: Matches `audit.key=ssh_authorized_keys` writes to `authorized_keys` (excluding `sshd`) on `ubuntu-01` .
`100204`	**Remote Desktop (RDP) login from external IP**	`T1021.001` Lateral Mvmt (`TA0008`)	Lvl 8 **P3**	**M. Alodeh** *Rev: M. Bustami*	`2026-10-19T18:54:54.754Z` — Detects RemoteInteractive RDP (`Event 4624` , `LogonType=10`) from non-loopback IP (`192.168.99.1` → `Win10`).
`100205`	**Suspicious rundll32 proxy execution**	`T1218.011` Stealth (`TA0005` v19)	Lvl 9 **P3**	**M. Alodeh** *Rev: S. Asender*	`2026-10-19T18:43:33.350Z` — Mapped to ATT&CK v19 **Stealth (TA0005)**. Fired on `rundll32.exe url.dll,FileProtocolHandler calc.exe` .
`100206`	**Registry Run/RunOnce key via SetValue**	`T1547.001` Persistence (`TA0003`)	Lvl 9 **P3**	**M. Alodeh** *Rev: M. Bustami*	`2026-10-19T18:43:34.422Z` — Detects Sysmon EID 13 `SetValue` on `\CurrentVersion\(Run|RunOnce)\` . Inherited from Sysmon rules `92300-92306` .
`100207`	**Windows Security/System event log cleared**	`T1685.005` Defense Impair (`TA0112`)	Lvl 12 **P1**	**M. Bustami** *Rev: M. Alodeh*	`2026-10-19T18:52:16.445Z` — Mapped to ATT&CK v19 **Defense Impairment (TA0112)**. Fired live on `wevtutil.exe cl Security` (`Event 1102`).

3. TECHNICAL ARCHITECTURE, TRIAGE SEPARATION OF DUTIES & REPOSITORY MAP

3.1 Live 3-VM Infrastructure & Three Integrated Log Sources (Sub-task 1.1)

VM NAME & HOSTNAME	IP ADDRESS (VMNET1)	OS & ROLE	WAZUH AGENT ID & STATUS	INTEGRATED LOG SOURCES & DECODERS	MEASURED DELAY
`SOC-Wazuh` (`wazuh.manager`)	`192.168.99.10`	Ubuntu 24.04 LTS + Docker Engine (`wazuh-docker` v4.14.9 Single-Node)	`ID: 000` **ACTIVE / LOCAL**	Wazuh Manager (`analysisd` , `remoted`), OpenSearch Indexer (`wazuh-alerts-*`), OpenSearch Dashboards (`HTTPS :443`)	Local Real-Time
`SOC-Win10` (`Win10`)	`192.168.99.30`	Windows 11 Enterprise Eval (Build 26100) — Clinical Admin Endpoint	`ID: 002` **ACTIVE**	**Log Source #1**: `Microsoft-Windows-Sysmon/Operational` (Sysmon v15.15 with patched `SwiftOnSecurity` config), Windows `Security` (4624, 4625, 4688, 1102), `System` (104), `PowerShell/Operational` (4104) via `windows_eventchannel`	**0.57 sec**
`SOC-ubuntu-01` (`ubuntu-01`)	`192.168.99.40`	Ubuntu 24.04 LTS — Patient Portal Web Server & Batch Host	`ID: 001` **ACTIVE**	**Log Source #2**: Linux `auditd` (`/var/log/audit/audit.log` with tuned `Neo23x0` + `10-soc-capstone.rules`) & `/var/log/auth.log` **Log Source #3**: `nginx` Web Server (`/var/log/nginx/access.log` & `error.log` via `web-accesslog` decoder)	**0.83 sec** (auditd) **0.45 sec** (nginx)

3.2 Alert Triage Tickets & Red/Blue Separation of Duties (Sub-tasks 1.5 & 1.6)

TICKET ID & FILE	INCIDENT CLASSIFICATION	RED-TEAM TRIGGER / SOURCE	ASSIGNED BLUE-TEAM ANALYST	PEER REVIEWER	KEY CONTAINMENT & FORENSIC OUTCOME
`SOC-2026-INC-A01` `triage/ticket_01_incident_a.md`	**P1 CRITICAL** **True Positive** (8-Rule Multi-Stage Chain on `Win10`)	**Saif Asender** (Blind Attack Chain Simulation)	**Mohammad Bustami** (Lead Blue-Team Investigator)	Mohammad Alodeh	6-Field Escalation Call placed; `Win10` isolated via host firewall (preserving `192.168.99.10:1514`); VMware snapshot `SOC-Win10-Snapshot1.vmsn` + 4 GB RAM image hashed in `SHA256SUMS.txt` ; full report in `INCIDENT_A_REPORT.md` .
`SOC-2026-INC-B02` `triage/ticket_02_incident_b_fp.md`	**P4 INFO / FP** **False Positive** (Rule `100201` LSASS Memory Access Noise)	OS Baseline Activity (`svchost.exe` & `MsMpEng.exe 0x1000`)	**Mohammad Alodeh** (Lead FP & Tuning Analyst)	Mohammad Bustami	Verified `grantedAccess=0x1000` under Windows 11 `RunAsPPL` from signed system binaries; tuned PCRE2 lookahead in `local_rules.xml` ; reduced false positives from 26 alerts / 5m to **0**.
`SOC-2026-INC-C03` `triage/ticket_03_linux_ssh_persistence.md`	**P2 HIGH** **True Positive** (Rule `100202` SSH Key & Rule `100203` Cron Job on `ubuntu-01`)	**Saif Asender** (SSH Key & Cron Persistence)	**Mohammad Alodeh** (Tier-1 SOC Analyst)	Saif Asender	Traced `audit.auid=1000` (`soc`), removed unauthorized SSH key from `/home/kali/.ssh/authorized_keys` , purged rogue crontab, and restricted `/etc/cron.allow` .

TICKET ID & FILE	INCIDENT CLASSIFICATION	RED-TEAM TRIGGER / SOURCE	ASSIGNED BLUE-TEAM ANALYST	PEER REVIEWER	KEY CONTAINMENT & FORENSIC OUTCOME
`SOC-2026-INC-D04` `triage/ticket_04_defense_impairment.md`	**P1 CRITICAL** **True Positive** (Rule `100207` Windows Security Event Log Cleared)	`wevtutil.exe cl Security` (Event ID `1102` on `Win10`)	**Saif Asender** (Tier-1 SOC Analyst)	Mohammad Bustami	Confirmed all pre-wipe Windows Security events (`4625` , `4624` , `4688`) were safely preserved off-host in `SOC-Wazuh` within `0.57s` before `wevtutil` executed.

3.3 Complete Repository File & Deliverable Inventory (`https://github.com/MB-Cyber1/Wazuh-Capstone`)

DIRECTORY / MODULE	COMMITTED FILES	RUBRIC MAPPING & SQUAD OWNERSHIP
Root & Governance	`README.md` , `CONTRIBUTION_MAP.md` , `.gitignore`	Master index, 3-member ownership matrix (M. Bustami, S. Asender, M. Alodeh)
`lab/` *(Infrastructure & Configs)*	`lab/wazuh/deploy_wazuh.sh` , `lab/ubuntu-01/setup_ubuntu01.sh` , `lab/vm-builder/build_vms.py` , `lab/vm-builder/build_win_vm.py`	Sub-task 1.1 (6 pts): Automated 3-VM deployment, Sysmon, `auditd` , and `nginx` configuration
`detections/sigma/` *(10 Validated Sigma Rules)*	`01_100005_t1059_001_powershell_encoded.yml` through `10_100207_t1685_005_event_log_cleared.yml`	Sub-task 1.3 (12 pts): Sigma v2 rules validated with `sigma-cli` (0 errors)
`detections/wazuh/` **&** **`CONVERSION_LOG.md`**	`detections/wazuh/local_rules.xml` , `detections/CONVERSION_LOG.md`	Sub-tasks 1.2 & 1.3: Production Wazuh XML rules (`100005-100207`) & manual conversion log
`detections/cards/` *(10 Detection Cards)*	`card_01_rule_100005.md` through `card_10_rule_100207.md`	Sub-tasks 1.4 & 2.2: One-page detection cards peer-reviewed and aligned with Part 2 P1–P4 SLAs
`detections/tuning/` , **`navigator/`** , **`dashboard/`**	`TUNING_REPORT.md` , `attack_navigator_v19.json` , `GAP_NOTE.md` , `soc_capstone_tier1_dashboard.ndjson` , `DASHBOARD_GUIDE.md`	Sub-tasks 1.3 & 1.4: Before/after tuning counts, ATT&CK v19 Navigator layer, 6-panel SIEM dashboard
`evidence/` *(Live SIEM Proof & Hashes)*	`PIPELINE_PROOF.md` , `RULE_EVIDENCE_SUMMARY.md` , `alerts_evidence.json` , `SHA256SUMS.txt`	Sub-tasks 1.1–1.5: Sub-second latency proof, raw fired alert JSONs, VM snapshot SHA-256 hashes
`triage/` *(Tickets, Escalation & Report)*	`ticket_01_incident_a.md` .. `ticket_04_defense_impairment.md` , `escalation_call_incident_a.md` , `containment_and_evidence_preservation.md` , `INCIDENT_A_REPORT.md`	Sub-tasks 1.5 & 1.6 (23 pts): 4 complete 10-item tickets, 6-field call, host containment, Incident A full report
`readiness/` *(Part 2 Readiness Pack)*	`01_org_profile_and_risk_register.md` through `08_executive_briefing.md`	Sub-tasks 2.1 to 2.8 (40 pts): Complete Incident Readiness Pack, Cold Test log, and TTX Observer timings

GitHub Repository: `https://github.com/MB-Cyber1/Wazuh-Capstone`

Squad: Mohammad Bustami • Saif Asender • Mohammad Alodeh

Live Wazuh SIEM: `https://192.168.99.10`